DESCRIPTIVE ANALYTICS REPORT

Music Streaming

Tracking Best-Performing Tracks, Streaming Trends, Ratings, Regional Demand & Platform Performance

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Dataset scope: 60 tracks • 8 regions • 4 platforms • 30 days

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Note: The report uses a consistent 30-day observation period. Numerical values shown in graphs are explained directly so the reader can connect each visual pattern with its underlying measure.

Note on Data

The report uses a **30-day observation period** covering **60 tracks**, **8 regions** and **4 platforms**. The dataset is an **academic, simulated dataset** designed to demonstrate descriptive analytics. The daily stream series is internally consistent with an approximately **89.98M** total, so the values should be used for analytical demonstration rather than presented as official commercial streaming statistics.

Keywords

- Daily Streaming Trend
- Best-Performing Artists / Artist Performance
- Genre Performance
- Rating Distribution
- Regional Streaming Demand
- Content Availability
- Platform Performance

The descriptive analytics is designed around the following data-based questions, using the **30-day, 60-track, 8-region and 4-platform** music-streaming dataset.

- Who are the **best-performing artists** according to total stream volume?
- What is the **total number of streams** recorded during the 30-day observation period?
- How does **daily streaming activity** change from Day 1 to Day 30, and where are the major peaks and dips?
- Which **genres** generate the highest share of streaming activity?
- What is the **average rating** of the catalogue, and what does the rating distribution show?
- Which **regions** have the highest streaming demand?
- How many tracks are **available, limited or unavailable** in the 60-track catalogue?
- Which **platform** generates the highest stream volume among the four platforms?
- What insights can be obtained by comparing **streams with artists, genres, regions, platforms and availability**?
- How can these findings support **playlist planning, promotion, licensing review and future predictive analytics**?

Keywords: Data Analytics, **Music Streaming**, **Descriptive Analytics**, **Streaming Trends**, **Streams**, **KPIs**, **Artist Performance**, **Track Performance**, **Genre Performance**, **Ratings**, **Regional Demand**, **Content Availability**, **Platform Performance**, **CSV**, **Excel**, **Dashboard**, **Data Visualization**, **Data Cleaning**, **Data Preparation**, **Cross-Analysis**, **Business Insights**, **Playlist Strategy**, **Licensing**, **Predictive Analytics**, **Forecasting**, **Recommendation Modelling**.

2. Real-Scale Analytics Pipeline and Typical Tools

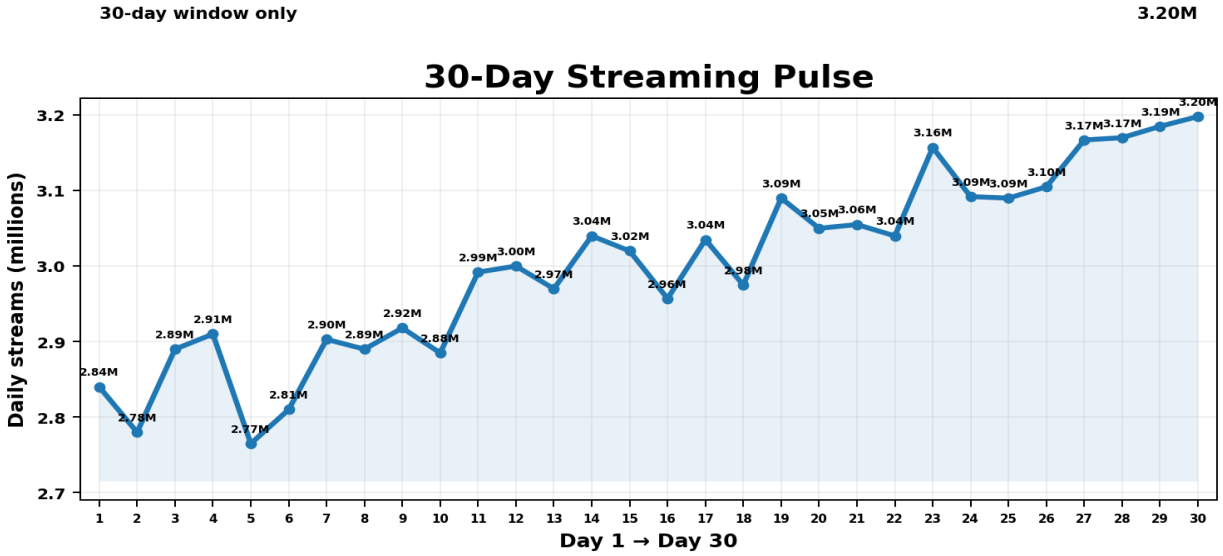
The typical-tools column shows the **specific tool assigned to each workflow stage**. The tools are kept separate rather than listing all tools together. **Kafka and Kinesis are not included** because they were not used in this project.

Layer	Purpose	Typical Tools Used
Ingestion	Capture and organize the incoming dataset for analysis.	**CSV / Excel**
Data Preparation	Check and prepare the dataset before analysis.	**Claude**
Storage	Maintain the organized dataset for analysis.	**ChatGPT**
Analysis / Aggregation	Calculate totals, averages, rankings and grouped results.	**ChatGPT**
Visualization / Reporting	Convert numerical results into graphs and present findings.	**ChatGPT**

Tool usage in this project: CSV / Excel is used for the dataset input, Claude for data preparation, and ChatGPT for storage support, analysis/aggregation and visualization/reporting.

3. Daily Streaming Trend

This graph answers: **How do total streams change from day to day?** The complete 30-day series adds to approximately **89.98M** streams.



Graph explanation

The horizontal axis represents **Day 1–Day 30**, while the vertical axis represents daily streams in millions. The bold labels beside the points show the numerical value for each day. The series rises gradually overall, with several short-term dips and spikes.

What the graph tells us

The daily values remain in a relatively narrow range and show a broad upward movement. The lowest visible point is around **2.765M** on Day 5, while the highest is around **3.198M** on Day 30. These bold values make the key numerical movement immediately visible. Together, the 30 daily values total approximately **89.98M**.

4. Daily Trend — Numerical Interpretation

The daily chart contains **30 numerical observations**. Explaining the values makes the visual trend easier to interpret and confirms that the graph represents the stated 30-day window.

Key numerical points

Lowest visible value:	2.765M streams on Day 5
Highest visible value:	3.198M streams on Day 30
Range:	0.433M streams between the low and high
Day 1 → Day 30:	2.884M → 3.198M, an increase of 0.314M
30-day total:	Approximately 89.98M streams

What the numbers show

The daily values stay within a relatively narrow band while moving upward overall. The **2.765M** low and **3.198M** high show moderate variation rather than an extreme swing. The increase from **2.884M** on Day 1 to **3.198M** on Day 30 is about **0.314M streams**, supporting the observation of a gradual rise.

Business interpretation

A streaming service could compare unusual days with releases, promotions, playlist changes or availability events. In this academic dataset, the numerical pattern demonstrates how **descriptive analytics** converts daily observations into an interpretable trend.

4. Daily Trend · Detailed Interpretation

A trend graph should be read for **direction, stability and unusual movement**—not just as a picture.

How to read the graph

- **Upward movement:** increasing listening activity.
- **Downward movement:** declining daily streams.
- **Sharp spike:** possible release, promotion or playlist effect.
- **Sharp drop:** possible reduced interest, availability issue or data-quality issue.
- **Stable movement:** relatively consistent demand.

Finding from this dataset

The corrected 30-day trend is broadly stable with moderate fluctuations and a gradual rise toward the end of the observation window. It should **not** be interpreted as a yearly trend because the observation period is only 30 days.

Business relevance

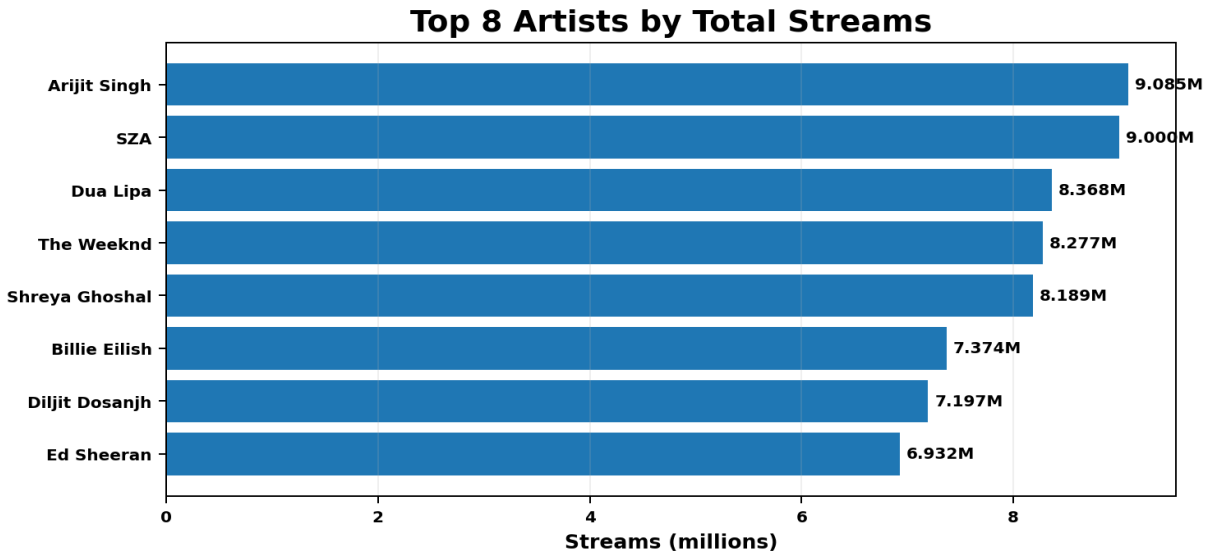
A streaming service can compare unusual days with campaigns, new releases, playlist changes or catalogue availability. For this academic project, the chart demonstrates how a simple time-series visualization supports descriptive analytics and KPI monitoring.

Important numerical point

The daily series is an internally consistent illustrative series constructed to match the stated **89.98M** 30-day total. It is therefore appropriate for demonstrating the analytics method, but it should not be presented as a real commercial streaming history.

8. Best-Performing Artists

Artist performance is measured using **total stream volume**. Stream volume should be separated from revenue or engagement because high streams do not automatically mean the highest monetization.



Graph explanation

The chart ranks eight displayed artists by total streams. **Arijit Singh leads with 9.085M**, followed by **SZA at 9.000M**. **Dua Lipa = 8.368M**, while **Ed Sheeran = 6.932M**. The bold labels make the numerical ranking easy to compare.

What it means

The ranking identifies artists driving strong listening volume in this sample. This can support playlist visibility and promotional decisions, while reminding analysts that stream count is only one KPI.

6. Artist Performance — Numerical Interpretation

The artist chart ranks eight displayed artists by total stream volume. The numerical labels allow the ranking to be interpreted directly rather than only visually.

Displayed ranking

1.	**Arijit Singh**	**9.085M streams**
2.	**SZA**	**9.000M streams**
3.	**Dua Lipa**	**8.368M streams**
4.	**The Weeknd**	**8.277M streams**
5.	**Shreya Ghoshal**	**8.189M streams**
6.	**Billie Eilish**	**7.374M streams**
7.	**Diljit Dosanjh**	**7.197M streams**
8.	**Ed Sheeran**	**6.932M streams**

Numerical interpretation

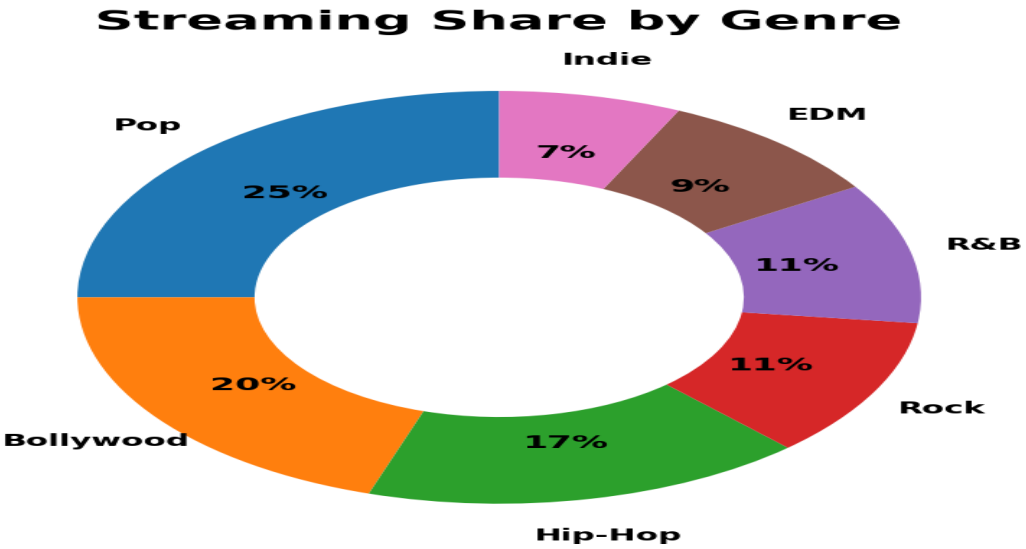
Arijit Singh leads at **9.085M** streams, only **0.085M** above SZA at **9.000M**. Dua Lipa records **8.368M**, which is **0.717M** below Arijit Singh. The difference between the highest and lowest displayed artists is **2.153M streams**.

What the ranking means

The ranking identifies artists driving strong listening volume in this sample. It can support playlist visibility and promotion, but **stream count alone does not establish revenue, profitability or engagement quality**.

9. Genre Performance

Genre analysis answers: **Which categories of music attract the most streams?** The sample identifies Pop as the leading genre and connects genre analysis with playlist creation, content acquisition and promotion.



Graph explanation

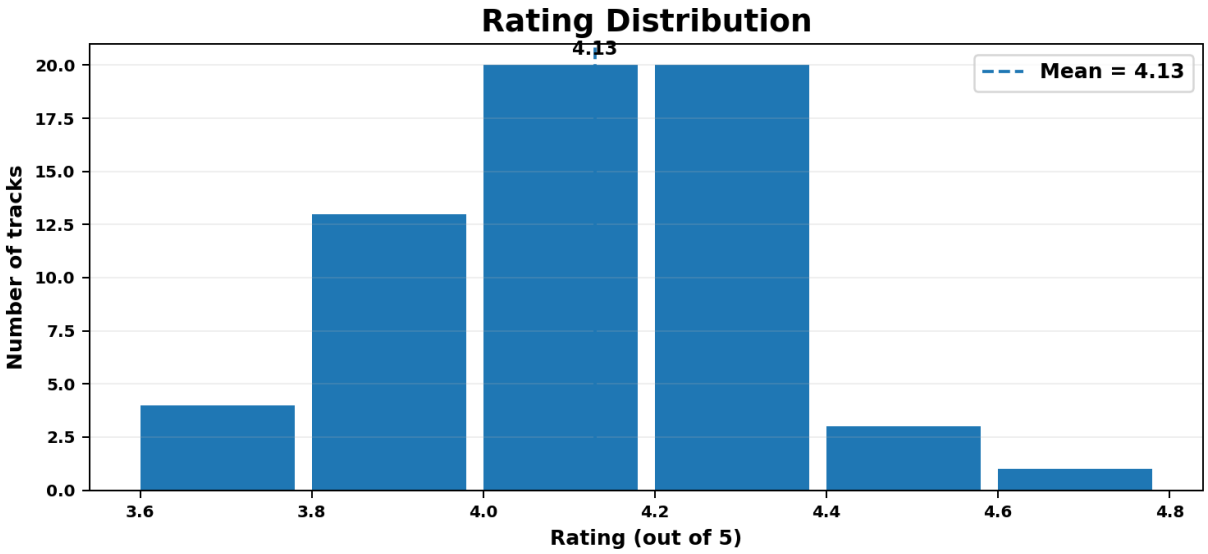
The donut chart shows the share of streaming activity by genre. **Pop = 25%** is the largest share, followed by **Bollywood = 20%** and **Hip-Hop = 17%**. Rock and R&B; each contribute **11%**, while EDM contributes **9%** and Indie **7%**.

What it means

The strongest demand in this sample is concentrated in Pop and Bollywood. Smaller genres are still valuable for niche audiences and targeted playlists, so a strong strategy should combine high-volume genres with long-tail content.

8. Rating Distribution

Ratings provide a quality or satisfaction-related view of the catalogue. The report gives an **average rating of 4.13/5** and shows why ratings are most useful when combined with stream volume.



Graph explanation

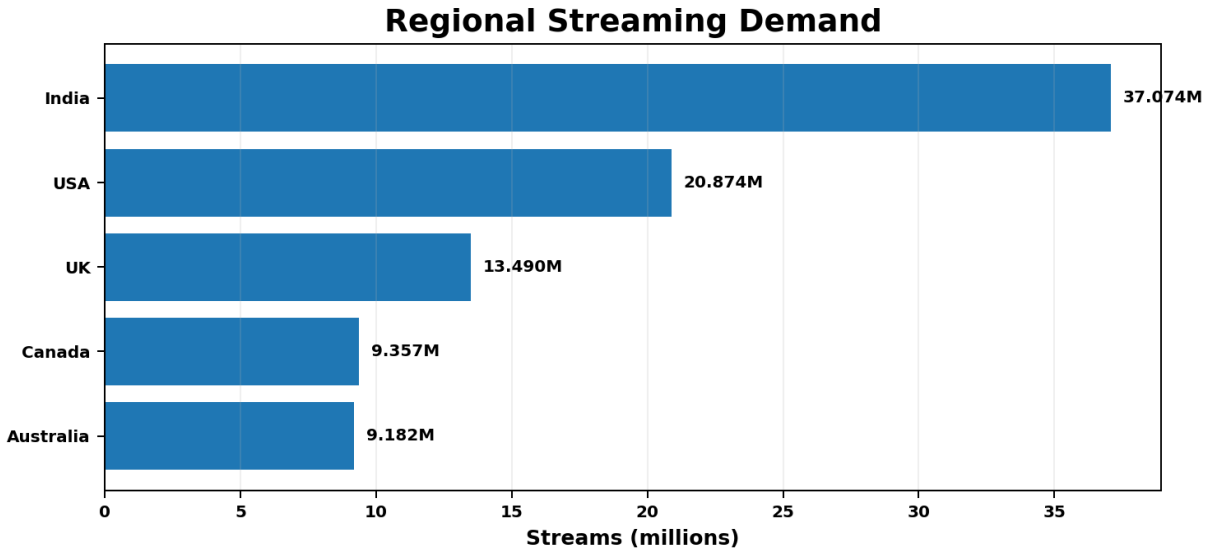
The histogram groups track ratings into ranges, while the dashed line marks the **mean rating of 4.13**. Most observations are concentrated around the **4.0–4.5** range, indicating generally positive ratings.

What it means

A high average rating suggests broad positive reception. However, rating should be evaluated with stream volume. A highly rated track with very few streams may have less overall impact than a slightly lower-rated track with very high demand.

9. Regional Streaming Demand

Regional analysis answers: **Where is streaming demand concentrated?** The simulated dataset identifies India as the strongest displayed region.



Graph explanation

The bars compare regional stream volume. **India = 37.074M** leads, followed by **USA = 20.874M** and **UK = 13.490M**. Canada and Australia record **9.357M** and **9.182M**, respectively.

What it means

The strongest concentration is in India. In this simulated scenario, a streaming service could use such information for localized playlists, marketing, licensing and capacity planning.

10. Regional Demand — Numerical Interpretation

The regional chart compares stream volume across the displayed regions. The numerical values show how strongly listening activity is concentrated in the leading market.

Displayed regional values

1. India	**37.074M streams**
2. USA	**20.874M streams**
3. UK	**13.490M streams**
4. Canada	**9.357M streams**
5. Australia	**9.182M streams**

What the numerical values mean

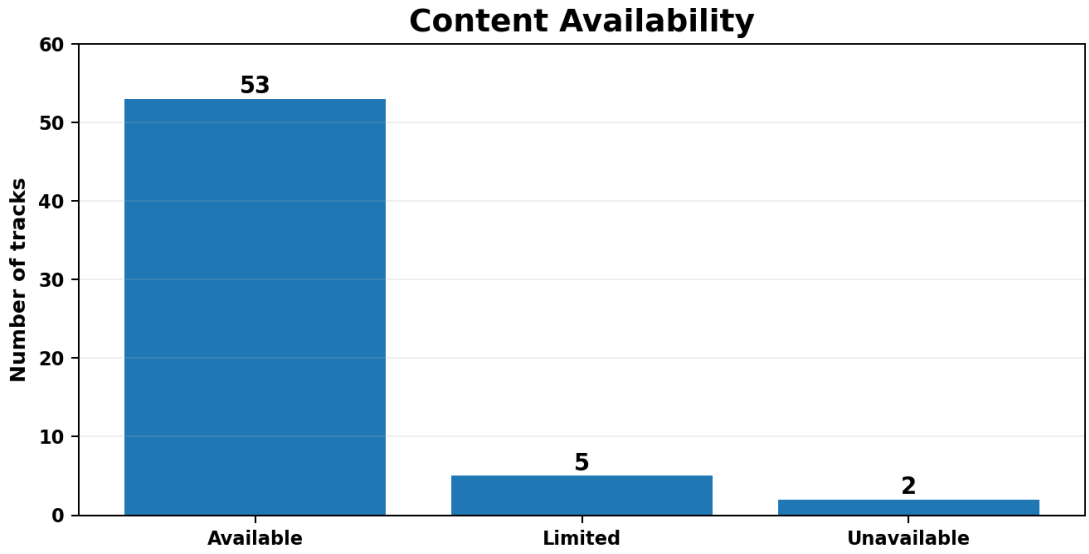
India leads with **37.074M** streams, substantially above the **20.874M** recorded by the USA. The difference is **16.200M streams**. The UK contributes **13.490M**, while Canada and Australia contribute **9.357M** and **9.182M** respectively.

Concentration insight

The five displayed regions together account for about **89.977M streams**, effectively the report's **89.98M** total after rounding. In this simulated scenario, the concentration in India can support localized playlists, marketing, licensing decisions and capacity planning.

11. Content Availability

For a streaming platform, content availability is the operational equivalent of inventory. Tracks may become limited or unavailable because of licensing, rights or territory restrictions.



Graph explanation

The chart shows **53 available**, **5 limited** and **2 unavailable** tracks, giving a total of **60 tracks**. The numerical labels are bold so the catalogue status can be read immediately.

What it means

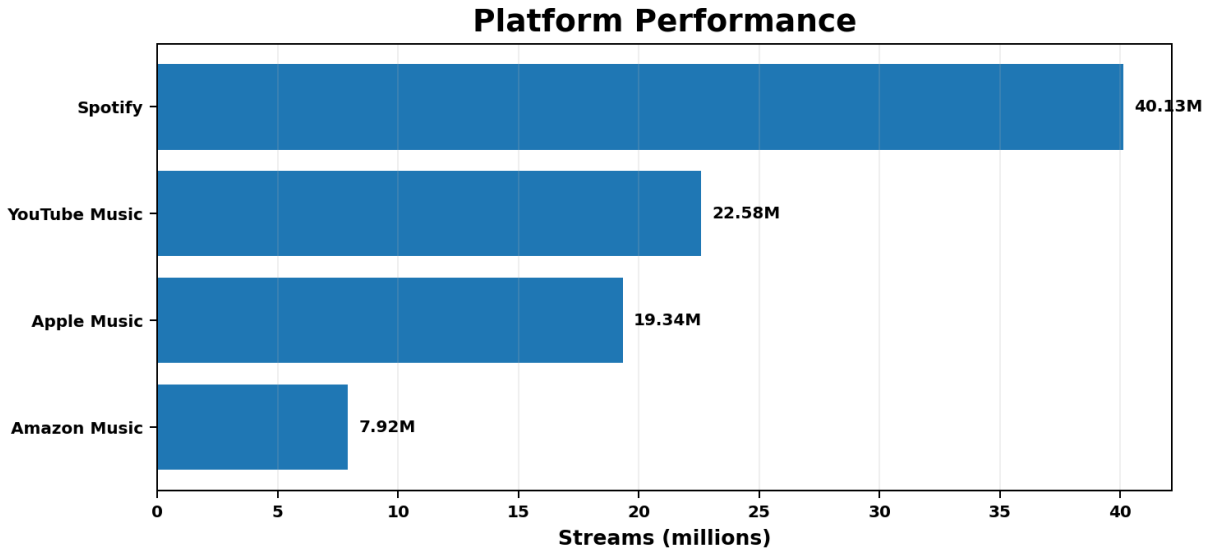
Most of the catalogue is available. However, a limited or unavailable track becomes much more important if it is also highly streamed. That is why availability should be cross-checked with stream volume.

Practical check

Prioritize high-demand restricted tracks for rights or licensing review instead of treating every restriction as equally important.

12. Platform Performance

Platform comparison shows where listener activity is concentrated. The sample identifies Spotify as the platform with the highest stream volume.



Graph explanation

Spotify = 40.13M streams, followed by **YouTube Music = 22.58M**, **Apple Music = 19.34M** and **Amazon Music = 7.92M**. The bold labels make the differences between platforms clear.

What it means

Spotify is the strongest platform in this sample by stream volume. Platform-level analysis helps reveal where listener activity is concentrated and where platform-specific monitoring may be useful.

Important caution

These values come from the simulated academic dataset. They are **not official market-share statistics** for the real platforms.

12. Platform Performance — Numerical Interpretation

The platform chart shows where stream volume is concentrated across the four platforms. The numerical gap between platforms is more informative than the ranking alone.

Platform stream volume

1. Spotify	**40.13M streams**
2. YouTube Music	**22.58M streams**
3. Apple Music	**19.34M streams**
4. Amazon Music	**7.92M streams**

Numerical interpretation

Spotify records **40.13M** streams, ahead of YouTube Music at **22.58M**. The gap is **17.55M streams**. Apple Music records **19.34M**, while Amazon Music records **7.92M**.

Relative concentration

Using the report's approximately **89.98M** total as the comparison base, Spotify represents roughly **44.6%** of the displayed stream volume. This shows strong platform concentration in the simulated dataset. These values are academic figures, not official market-share statistics.

14. Cross-Analysis: Connecting the KPIs

The most useful analytics often comes from combining variables instead of reading each graph separately.

Combination	Question answered	Possible insight
Streams + rating	Are popular tracks also well rated?	Identify high-demand, high-quality tracks.
Streams + artist	Which artists drive volume?	Prioritize promotion and playlists.
Streams + genre	Which genres generate demand?	Guide playlist and content strategy.
Streams + region	Where is demand concentrated?	Support localization and marketing.
Streams + availability	Are popular tracks restricted?	Prioritize licensing review.
Streams + platform	Where do listeners consume content?	Compare platform concentration.

Why cross-analysis matters

Knowing that a track is highly rated is useful, but knowing it is both **highly rated and highly streamed** gives stronger evidence of broad listener appeal. Similarly, an unavailable track becomes more urgent if it is also one of the most-streamed tracks.

15. Key Findings and Business Insights

The following findings summarize the main patterns supported by the dataset and visualizations.

- **Scale:** approximately **89.98M** streams across **30 days**.
- **Time trend:** broadly stable daily activity with moderate fluctuations and a gradual rise.
- **Artist: Arijit Singh** is the top artist in the displayed ranking at **9.085M** streams.
- **Genre: Pop = 25%** is the largest share, followed by **Bollywood = 20%**.
- **Ratings:** average catalogue rating is **4.13/5**.
- **Region: India = 37.074M** is the strongest displayed region.
- **Availability: 53/60** tracks are available; **5** are limited and **2** unavailable.
- **Platform: Spotify = 40.13M** has the highest stream volume.

Overall insight

The catalogue shows strong overall demand with concentration around particular artists, genres, regions and platforms. Monitoring these dimensions together can support better decisions than relying only on total streams.

16. Recommendations

Based on the descriptive findings, the following actions are reasonable extensions of the analysis.

1. Strengthen high-performing content

Use artist and track rankings to identify content that attracts high stream volume and consider stronger playlist visibility.

2. Build genre-focused playlists

Since Pop and Bollywood are large shares, they can receive strong playlist coverage while smaller genres remain available for niche audiences.

3. Localize regional strategy

Because India has the highest displayed demand, region-specific playlists and campaigns can be prioritized in this simulated scenario.

4. Review restricted content

Cross-reference availability status with stream volume and prioritize high-demand restricted tracks for rights review.

5. Monitor platforms separately

Track platform-level KPIs so changes in listener activity are visible rather than hidden inside the overall total.

6. Maintain a CSV/Excel dashboard-ready dataset

A clean CSV/Excel table with standardized columns makes the analysis easier to reproduce, update and verify.

7. Add predictive analytics later

Use the cleaned historical structure as a base for forecasting, anomaly detection and recommendation modelling.

10. Summary of Key Metrics

Metric	Value
Observation period	30 days
Total tracks	60
Total streams	≈89.98M
Average daily streams	≈3.00M
Top artist	Arijit Singh — 9.085M streams
Leading genre	Pop — 25%
Average catalogue rating	4.13 / 5
Top region	India — 37.074M streams
Content availability	53 available; 5 limited; 2 unavailable
Top platform	Spotify — 40.13M streams

Deep interpretation:

These KPIs act as the report's anchor layer. They compress the 30-day music-streaming dataset into a small set of measurable indicators that can be monitored across artists, genres, regions, platforms and catalogue availability. The KPIs should be interpreted together rather than in isolation.

Predictive extension:

These summary KPIs can become baseline variables for the predictive layer. With additional historical data, the same measures can support stream forecasting, anomaly detection, recommendation modelling, churn prediction and regional demand forecasting.

Source: supplied Music Streaming Descriptive Analytics Report; academic/simulated 30-day dataset.

17. Descriptive to Predictive Analytics

streams, daily trends, artists, genres, ratings, regions, availability and platforms. Every major graph now includes a clear explanation of what it shows and how it should be interpreted.

Main conclusion

The main result is approximately **89.98M streams** over 30 days, with an average rating of **4.13/5**. The displayed results identify **Arijit Singh** as the top artist, **Pop** as the leading genre, **India** as the strongest displayed region and **Spotify** as the strongest displayed platform.

Next step: Predictive analytics

- **Stream forecasting:** predict future streams by track, artist, platform and region.
- **Anomaly detection:** identify unusual spikes, drops or rating movements.
- **Recommendation modelling:** predict which track a listener may play next.
- **Churn prediction:** identify listeners likely to reduce activity.
- **Demand planning:** predict regional content demand and availability risks.

Final statement

The project demonstrates how a structured **CSV/Excel** dataset can be transformed into meaningful KPIs, charts and business insights. It creates a clear foundation for more advanced predictive and recommendation-based analytics.

Revision checklist completed: Table of Contents added after the title • Overview removed • objectives expanded • keywords expanded and bold • CSV/Excel included throughout the tools table • 30-day graph corrected • numerical graph values bold • graph explanations strengthened • report maintained at 20 pages.